

Heart Disease Prediction: A Classification Model Comparison

I. Introduction

The primary objective of this report is to identify the most suitable classification model for predicting heart disease status using a dataset comprising various medical and physiological attributes. Early detection of heart disease can significantly improve patient outcomes, and statistical modeling offers a data-driven approach to assist in this endeavor.

The dataset contains 300 observations and 13 variables, where the target variable, disease, indicates the presence ("yes") or absence ("no") of heart disease. The input variables include:

- Age, blood pressure (bp), cholesterol level (chol), and maximum heart rate (heart.rate)
- Categorical indicators like sex, chest pain type (chest.pain), fasting blood sugar (fbs), and exercise-induced angina (angina)
- Additional measures like ST depression, number of major vessels, and resting ECG results

In the subsequent sections of the report, we will follow these steps:

1. **Data Exploration & Preprocessing** - Understand the data, check for association between the response and input variables, and remove insignificant variables if needed
2. **Model Selection** – Apply and compare several classification algorithms including:
 - Logistic Regression
 - K-Nearest Neighbors (KNN)
 - Decision Tree
3. **Model Evaluation** - Evaluate models using true-positive rate, precision, and AUC - ROC.
4. **Recommendation** - Conclude by proposing the best-performing classifier based on a balance of performance metrics and interpretability.

This structured approach will allow us to rigorously assess multiple classifiers and recommend the most appropriate model for predicting heart disease risk.

II. EDA: Exploring the variables and association

1. The response and input variables

a) Response variable

Out of 300 people in the data set, 138 people – 46% display the presence of heart disease. This rate is incredibly high, therefore does not reflect the real world.

b) Input variables

- **age** (numeric): Age of the patient in years.
- **sex** (binary: 0 = female, 1 = male): Gender may influence risk levels due to biological differences.
- **chest.pain** (categorical): Type of chest pain experienced (0 = typical angina, 1 = atypical angina, 2 = non-anginal pain, 3 = asymptomatic), which can indicate different types of cardiac issues.

Due to the complexity in determining the severity of typical and atypical angina, we will combine them into one group – anginal pain, and rank the groups as follows:

1 - non-anginal pain (has no impact on patients' mortality)

2 - anginal pain (shows clear sign of chest pain)

3 - asymptomatic (shows no clear symptoms even though they may have significant underlying heart disease, thus more dangerous)

- bp (numeric): Resting blood pressure. Elevated blood pressure is a known risk factor.

- chol (numeric): Serum cholesterol level in mg/dl. Higher levels may be associated with heart disease.

- fbs (binary: 0 = false, 1 = true): Fasting blood sugar > 120 mg/dl, indicating potential glucose-related issues.

- rest.ecg (categorical): Resting electrocardiographic results (0 = normal, 1 = ST-T wave abnormality, 2 = possible or definite left ventricular hypertrophy). Abnormal ECG results can be indicative of heart abnormalities.

- heart.rate (numeric): Maximum heart rate achieved during a stress test - a measure of cardiac efficiency.

- angina (binary: 1 = yes, 0 = no): Indicates whether exercise-induced angina was present.

- st.depression (numeric): ST depression induced by exercise relative to rest - a potential sign of ischemia.

- vessels (categorical): Number of major vessels (0 - 4) colored by fluoroscopy.

- blood.disorder (ordinal): Represents a grading of blood-related health issues 1 = normal, 2 = fixed defect, 3 = reversible defect; 0 = missing value).

Note: For the missing values, we will classify them to the modal category, which is 2

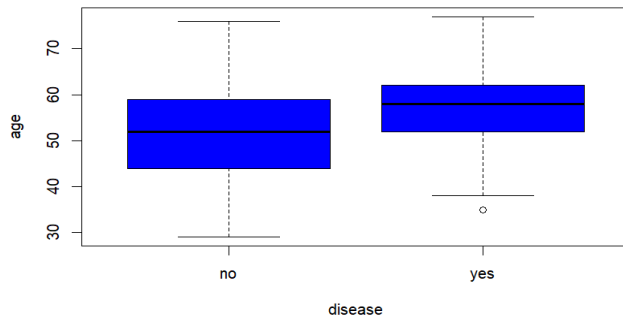
The ranking of blood.disorder by clinical severity in imaging results:

1 - Normal (High probability of functional recovery after revascularization)

2 - Reversible defect (Medium probability of functional recovery after revascularization)

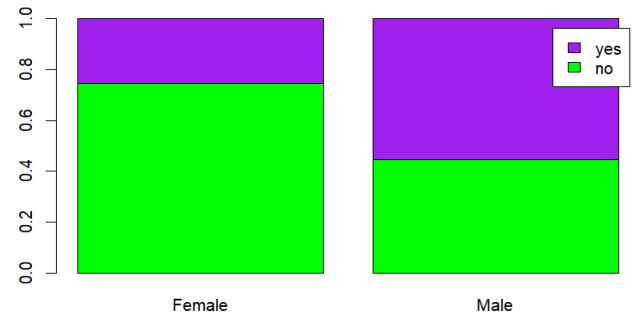
3 - Fixed defect (Low probability of functional recovery after revascularization)

2. The association between the response and each input variable



a) Age

This boxplot suggests that as age increases, the likelihood of heart disease increases, though there's some overlapping parts, meaning age alone is not a strong indicator



b) Sex

This bar plot indicates that males have a much higher chance of having disease than females, and the odds ratio of males having disease is 3.61 times that of females having disease, thus making it a strong indicator

c) Chest.pain

This bar plot suggests that people with anginal pain clearly have the highest chance of having heart disease, therefore being a strong indicator

d) bp

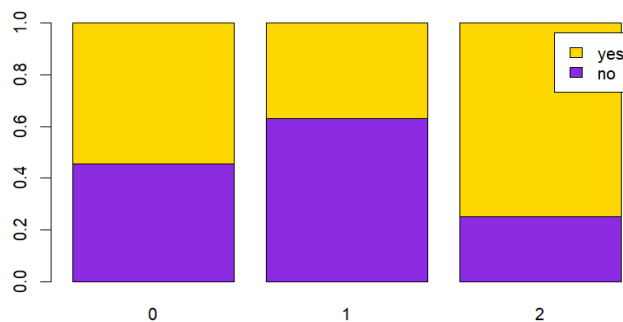
The boxplot shows similar medians and many overlapping parts, thus bp is not a significant indicator

e) chol

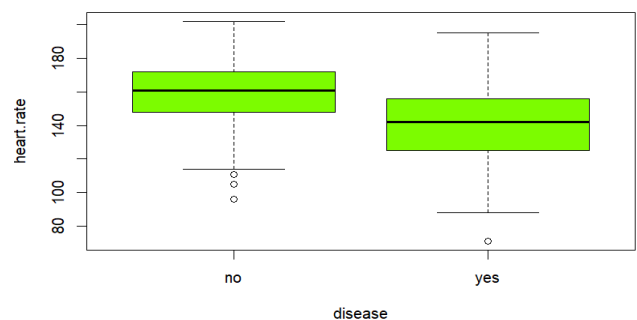
Like bp, the medians are similar and there are also many overlapping parts in this boxplot, suggesting that chol is an insignificant variable

f) fbs

The bar plot does not suggest any clear signs and the odds ratio between people having fbs greater than > 120 mg/dl with disease and that without disease is 1.146 times, an insignificant variable

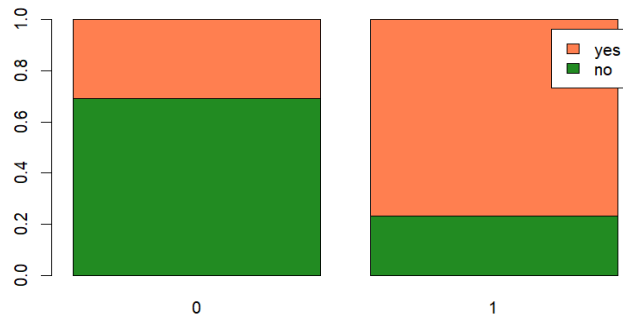


g) rest.ecg



h) heart.rate

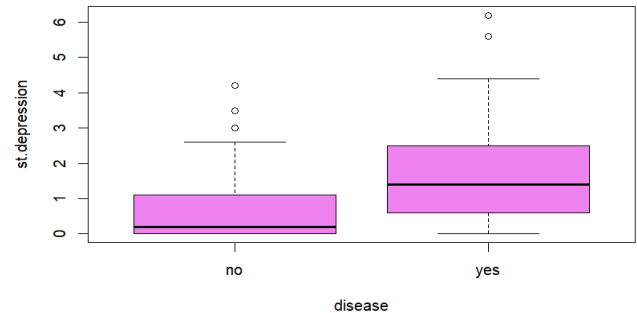
The bar plot shows that people with left ventricular hypertrophy have a high chance of heart disease, while it is not clear on the other two groups. There might be an association, but not certain



i) Angina

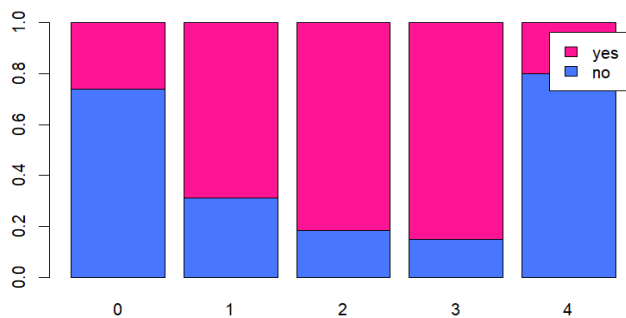
The bar plot indicates that people who have experienced angina induced by exercise have a much higher chance of heart disease, and the odds ratio of that is 7.41 times that of people who haven't experienced angina and have heart disease. This suggests a significant association with the response variable

The box plot shows a clear difference between the median of the people with and without heart disease, plus there is little overlapping between the two boxes, suggesting a strong relation.



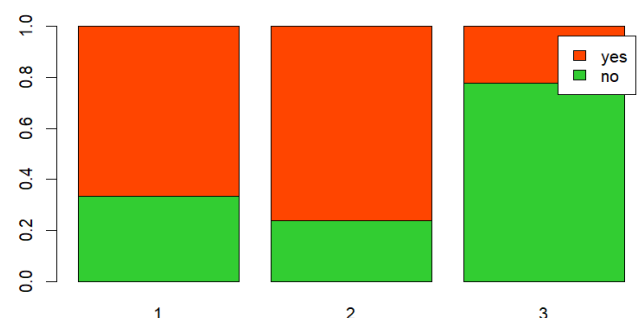
j) st.depression

We can see from the boxplot, the difference of medians between people with and without heart disease is significant, and the boxes have little overlapping parts. This association with the response variable is significant



k) vessels

The bar plot clearly shows that people with 0 or 4 major vessels have a low chance of getting heart disease, while the rest have a high probability. This suggests a clear association with the response variable



l) blood.disorder

The bar plot indicates that people with normal or reversible defect blood disorder have a high chance of having heart disease, and the opposite for people with fixed defect blood disorder. This shows a clear association with the response variable

Through this analysis, we can remove 3 variables from our data, which are bp, chol, and fbs as they show no association with our response variable

III. Methods: Building KNN, DT and LR Classifiers

1. Find the best version for each classifier

We will use 5-folds cross validation only for KNN and Decision Tree, as using it for Logistic Regression is too complicated.

Using TPR to determine the best k value for KNN, we get $k = 3$

Similarly, the best maxdepth for Decision Tree is $\text{maxdepth} = 2$

Finally, the inputs that produce the best Logistic Regression classifier includes sex, chest.pain, rest.ecg, heart.rate, angina, st.depression, vessels and blood.disorder (using a threshold of 0.4686)

2. Compare the performance of the classifiers with each other

a) TPR

KNN: 0.8406

DT: 0.8478

LR: 0.8623

b) Precision

KNN: 0.8992

DT: 0.8540

LR: 0.8623

c) ROC and AUC

KNN: 0.9601

DT: 0.8952

LR: 0.9307

3. The pros and cons of each classifier

1) KNN

Pros:

- Non-parametric: Makes no assumptions about data distribution - great for datasets like this with mixed types and potential nonlinear patterns.
- Simple and intuitive: Classifies based on “neighborhood” similarity.
- Can model complex boundaries between classes if k is chosen well.

Cons:

- Sensitive to feature scaling: Needs all features to be on the same scale
- Doesn't work well with categorical variables, especially significant ones such as angina, sex, even ordinal categorical variables such as blood.pressure, rest.ecg
- Performance drops in high dimensions.
- Computationally expensive at prediction time - especially with larger datasets.
- No model interpretability - it's a black-box decision based on distance.

2) Decision Tree

Pros:

- Interpretable structure: The tree format is easy to understand and explain to non-technical audiences.
- Handles both numerical and categorical data well.
- Captures non-linear relationships and interactions naturally.
- No need to scale data.

Cons:

- Prone to overfitting, especially on small or noisy datasets like this one.
- Unstable - small changes in data can result in a completely different tree.

3) Logistic Regression

Pros:

- Interpretable: Coefficients provide clear insight into how each variable affects heart disease risk.
- Well-suited to binary classification: Your target variable is binary (yes/no), which fits the LR model perfectly.
- Handles both numerical and categorical predictors: With proper encoding, LR performs reliably.
- Statistical diagnostics available: You can assess goodness-of-fit, confidence intervals, and significance of predictors.
- Performs well when the relationship is linear in the log-odds.

Cons:

- Assumes linearity in the log-odds of the predictors - which may not fully hold here.
- Less flexible for complex relationships between variables (e.g., interactions or non-linear boundaries).
- Sensitive to outliers and multicollinearity among features.

4. Propose the best classifier

Even though KNN has the highest score in terms of precision and AUC, the fact that most variables in our data are categorical variables, some are even binary variables such as angina and sex, with them being very significant, the distance calculation just doesn't make sense with these variables. On the other hand, Logistic Regression doesn't suffer from the same problem as KNN, and it still has a decent score in both precision and AUC, and it has the highest score in terms of TPR.

Therefore, after comparing the performance, as well as looking at the pros and cons of every classifier, we have ultimately decided that Logistic Regression is the best classifier.